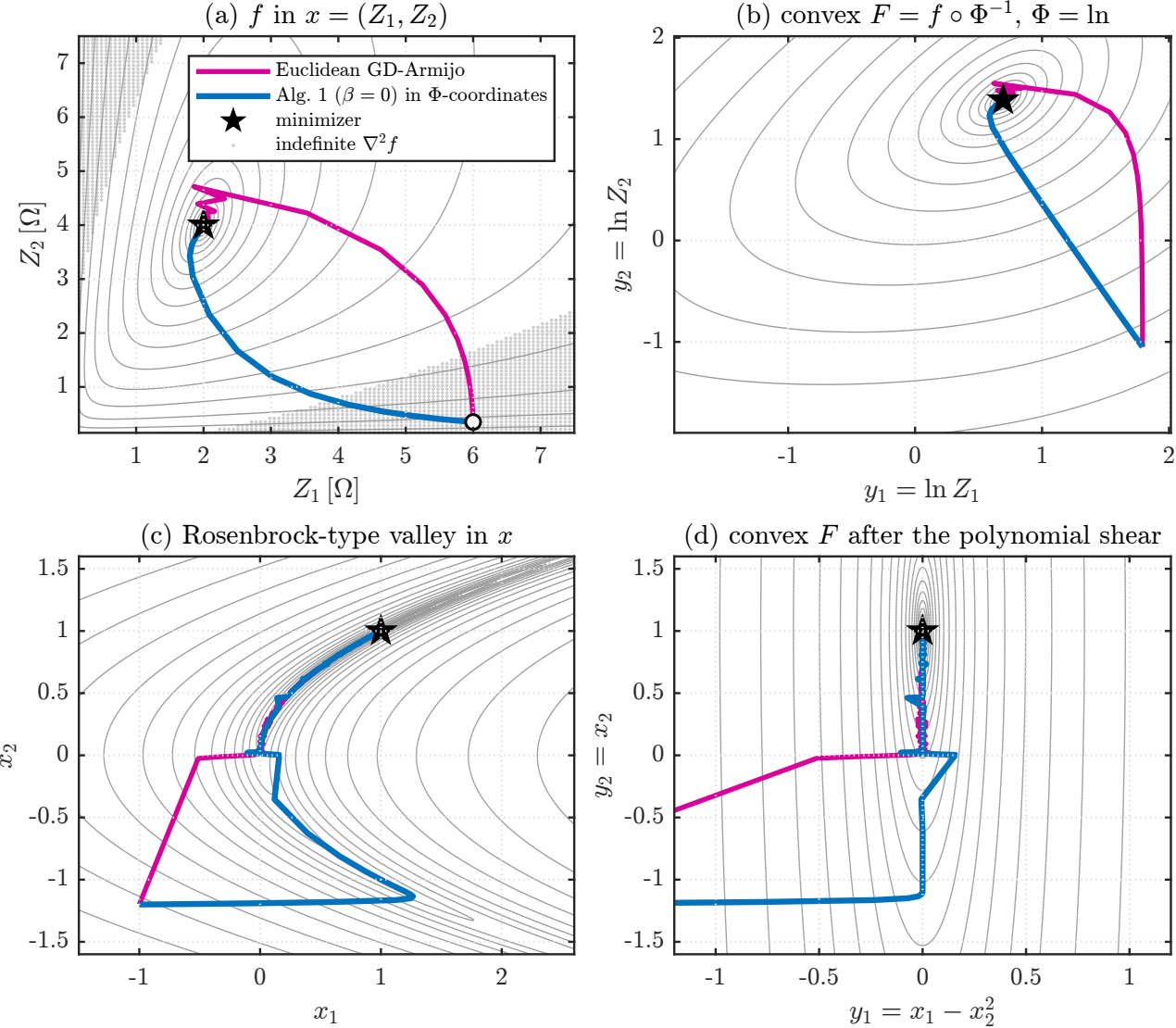

(a) f in $x = (Z_1, Z_2)$
Euclidean GD-Armijo
Alg. 1 ($\beta = 0$) in Φ-coordinates
minimizer
indefinite $\nabla^2 f$
$Z_2\ [\Omega]$
$Z_1\ [\Omega]$
(b) convex $F = f \circ \Phi^{-1}$, $\Phi = \ln$
$y_2 = \ln Z_2$
$y_1 = \ln Z_1$
(c) Rosenbrock-type valley in x
x_2
x_1
(d) convex F after the polynomial shear
$y_2 = x_2$
$y_1 = x_1 - x_2^2$